SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	18.08.2026	**Duration:**	60 minutes

Code of Conduct

I K R GAYANA (192471039)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: gayana k r__________________

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

1. Introduction & Scope

This document defines the test case design for the Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform. The platform integrates RFID, IoT sensors, AI-based storage optimization, and Autonomous Mobile Robots (AMRs) to automate inventory tracking, storage allocation, and order fulfillment while providing real-time visibility and performance reporting.

The purpose of this document is to identify the functional and non-functional requirements to be tested, define test scenarios derived from those requirements, and design detailed test cases — covering normal, boundary, invalid, and exceptional conditions — using recognized test design techniques. The document also provides a requirement-to-test-case traceability (coverage) matrix.

Scope: This test design covers the core modules of the platform — RFID-based item registration and scanning, real-time inventory updates, AI-driven storage allocation, discrepancy detection and alerting, order fulfillment and AMR routing, performance reporting, user authentication/access control, IoT environmental monitoring, and key non-functional characteristics (performance, scalability, security, usability, and reliability).

2. Requirements Identified for Testing

2.1 Functional Requirements (FR)

ID	Functional Requirement
FR-1	The system shall register and track inventory items using unique RFID tags.
FR-2	The system shall automatically update inventory counts in real time on inbound/outbound scans.
FR-3	The system shall use an AI algorithm to recommend/assign optimal storage locations based on item demand, size, and turnover rate.
FR-4	The system shall detect inventory discrepancies (scanned vs. recorded quantity) and generate alerts.
FR-5	The system shall support order fulfillment workflows (pick → pack → ship) with automated AMR task assignment and routing.
FR-6	The system shall generate on-demand and scheduled warehouse performance reports (throughput, accuracy, utilization).
FR-7	The system shall provide user authentication with role-based access control (Admin, Manager, Warehouse Staff).
FR-8	The system shall integrate with IoT sensors to monitor environmental conditions (temperature, humidity) where applicable.
FR-9	The system shall allow authorized personnel to manually correct/override inventory records with an audit trail.
FR-10	The system shall maintain a complete movement/audit history for every inventory item.

2.2 Non-Functional Requirements (NFR)

ID	Category	Non-Functional Requirement
NFR-1	Performance	RFID scan events shall reflect in inventory records within ≤ 2 seconds.
NFR-2	Scalability	The system shall support up to 500 concurrent RFID reader streams and 100,000+ SKUs without degradation.
NFR-3	Reliability / Availability	The system shall maintain ≥ 99.9% uptime and recover automatically from failures within the defined RTO.
NFR-4	Usability	A first-time user shall be able to perform core tasks (search, report generation) within 3 clicks / 1 minute.
NFR-5	Security	All data in transit/at rest shall be encrypted; access shall be restricted via role-based permissions and token-based API auth.
NFR-6	Accuracy	Inventory count accuracy shall be ≥ 99.5% at all times.
NFR-7	Interoperability	The system shall expose APIs to integrate with external ERP/WMS platforms.
NFR-8	Maintainability	The system shall be built on a modular architecture allowing independent updates to AI, RFID, and reporting modules.

3. Test Scenarios

Based on the requirements above, the following high-level test scenarios were identified. Each scenario is decomposed into detailed test cases in Section 5.

- TS-01: Registering new inventory items via RFID tags and validating tag uniqueness/format.
- TS-02: Real-time inventory quantity updates on inbound and outbound movements.
- TS-03: AI-driven storage location allocation for items of varying demand and size.
- TS-04: Detection and alerting of inventory discrepancies between physical scans and system records.
- TS-05: End-to-end order fulfillment workflow with AMR-based picking and routing.
- TS-06: Handling of exceptional conditions such as AMR faults, sensor failures, and network outages.
- TS-07: Generation and export of warehouse performance and utilization reports.
- TS-08: User authentication, session handling, and role-based access enforcement.
- TS-09: IoT-based environmental monitoring and threshold-based alerting.
- TS-10: System performance under concurrent RFID load and large data volumes.
- TS-11: Data security and API access control.
- TS-12: System recovery and data integrity after failure/downtime.

4. Test Design Techniques Applied

A combination of test design techniques was applied to ensure balanced and efficient coverage across valid, boundary, and invalid input spaces, as well as sequence-dependent (stateful) behaviour.

4.1 Equivalence Partitioning (EP)

Used to partition inputs such as RFID tag format, item demand classification (high/low turnover), and password length into valid and invalid classes, so that one representative test per partition is sufficient (e.g., TC-01, TC-03, TC-04, TC-12, TC-13).

4.2 Boundary Value Analysis (BVA)

Applied to numeric and length-bound fields — bin capacity, discrepancy threshold, and password length — by testing values at, just below, and just above the boundary.

Field	Min-1 (Invalid)	Min (Valid)	Nominal (Valid)	Max (Valid)	Max+1 (Invalid)
Bin Quantity (capacity = 100)	–	0	50	100	101
Password Length (8–20 chars)	7	8	14	20	21
Discrepancy Threshold (=3 units)	2 (no alert)	3 (alert)	–	–	–

4.3 Decision Table Testing

Applied to the inventory discrepancy alerting logic, where the alert action depends on the combination of quantity-variance level and system connectivity status.

Condition / Action	Rule 1	Rule 2	Rule 3	Rule 4
Scanned Qty vs System Qty	Equal	Within Tolerance	Exceeds Tolerance	Exceeds Tolerance
System Status	Online	Online	Online	Offline
Action: No Alert	✓	–	–	–
Action: Minor Variance Warning	–	✓	–	–
Action: Critical Discrepancy Alert	–	–	✓	–
Action: Queue Alert for Delivery on Reconnect	–	–	–	✓

4.4 State Transition Testing

Applied to the Order Fulfillment lifecycle (Pending → Picking → Packed → Shipped → Delivered, with a Cancelled branch), verifying both valid transitions and rejection of invalid/out-of-sequence transitions.

Current State	Event / Trigger	Next State	Valid?
Pending	Order created & stock available	Picking	Valid
Picking	AMR retrieves all items	Packed	Valid
Packed	Order handed to carrier	Shipped	Valid

Current State	Event / Trigger	Next State	Valid?
Shipped	Carrier confirms delivery	Delivered	Valid
Picking	Order cancelled by user	Cancelled	Valid
Pending	Order cancelled before picking starts	Cancelled	Valid
Shipped	Attempt to cancel after shipment	Cancelled	Invalid – must be rejected
Delivered	Attempt to re-enter Picking state	Picking	Invalid – must be rejected

5. Detailed Test Cases

The following test cases cover normal, boundary, invalid, and exceptional conditions for each module. 'Actual Result' and 'Pass/Fail Status' are left for entry during test execution.

Module 1: RFID Tag Registration & Scanning

Technique(s) applied: Equivalence Partitioning, Invalid/Exceptional Case Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-01	Register a new inventory item with a valid RFID tag	1. Login as Warehouse Staff. 2. Navigate to 'Register Item'. 3. Scan a new RFID tag on an item. 4. Enter item details and save.	RFID EPC: 24-char valid hex (e.g., 3034257BF400123456780001)	Item is registered successfully; RFID ID is linked to item record and appears in inventory list.	Not Executed	Pending
TC-02	Register item with RFID tag ID at maximum allowed length (boundary)	1. Scan/enter an RFID tag ID exactly at the max EPC length (96-bit / 24 hex chars). 2. Save record.	24-character hex string (boundary max)	System accepts the ID and stores it without truncation.	Not Executed	Pending
TC-03	Attempt to register an item using a duplicate RFID tag already in use	1. Scan an RFID tag already assigned to another item. 2. Attempt to save.	RFID ID already present in database	System rejects the operation and displays 'Duplicate RFID Tag' error; no new record created.	Not Executed	Pending
TC-04	Attempt to register item with a malformed RFID tag ID	1. Manually enter an RFID ID containing non-hex characters or incorrect length. 2. Attempt to save.	e.g., 'XYZ-123-###' (invalid format)	System rejects input and displays 'Invalid RFID format' validation error.	Not Executed	Pending
TC-05	RFID reader loses network connectivity mid-scan	1. Begin scanning items at a dock door. 2. Disconnect reader from network mid-operation. 3. Reconnect after 30 seconds.	Simulated network drop during active scan session	Scans performed during outage are buffered locally and synced automatically on reconnect; no data loss; user is alerted of offline mode.	Not Executed	Pending

Module 2: Real-Time Inventory Update

Technique(s) applied: Boundary Value Analysis, Exceptional Case Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-06	Outbound scan reduces inventory count correctly	1. Scan item at outbound gate. 2. Check inventory dashboard.	Item SKU with current qty = 50	Quantity updates to 49 within 2 seconds; movement logged with timestamp.	Not Executed	Pending
TC-07	Inbound scan increases inventory count correctly	1. Scan item at inbound receiving dock. 2. Check inventory dashboard.	Item SKU with current qty = 20	Quantity updates to 21 within 2 seconds; movement logged with timestamp.	Not Executed	Pending
TC-08	Outbound scan attempted when stock quantity is already zero (lower boundary)	1. Set item quantity to 0. 2. Scan the item for outbound movement.	Qty = 0 (boundary)	System blocks the movement, prevents negative stock, and raises a 'Stock Unavailable' alert.	Not Executed	Pending
TC-09	Inbound scan when storage bin is at maximum capacity (upper boundary)	1. Set bin capacity to max (e.g., 100 units). 2. Scan one additional unit inbound to the same bin.	Bin capacity = 100, current qty = 100	System rejects placement in that bin, triggers 'Bin Overflow' alert, and suggests an alternate bin.	Not Executed	Pending
TC-10	Same RFID tag read at two different locations within 1 second (conflicting scan)	1. Simulate two reader events for the same tag ID at different gates within 1 second.	Duplicate simultaneous scan event	System flags a 'Conflicting Read' exception, logs both events, and does not double-count the movement.	Not Executed	Pending
TC-11	Inventory sync delayed due to high network latency	1. Introduce artificial network delay (>5 sec) between edge reader and central server. 2. Perform a scan.	Simulated latency > 5000 ms	Scan event is queued locally and retried with exponential backoff; inventory eventually reflects correct count once synced; no duplicate entries.	Not Executed	Pending

Module 3: AI-Based Storage Allocation

Technique(s) applied: Equivalence Partitioning, Boundary Value Analysis

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-12	High-turnover item is allocated to a nearest pick location	1. Add new item flagged as 'fast-moving' (high historical demand). 2. Trigger AI storage allocation.	Item demand score in 'high' partition	AI assigns item to a bin within the primary pick zone (closest to dispatch).	Not Executed	Pending
TC-13	Low-turnover / bulk item is allocated to bulk/remote storage	1. Add new bulk item flagged as 'slow-moving'. 2. Trigger AI allocation.	Item demand score in 'low' partition, large volume	AI assigns item to bulk storage zone away from the primary pick face.	Not Executed	Pending
TC-14	Allocation requested	1. Simulate warehouse utilization =	Utilization = 100% (upper	System rejects allocation, returns 'No	Not Executed	Pending

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
	when warehouse storage is at 100% capacity (boundary)	100%. 2. Attempt to allocate a new item.	boundary)	Available Storage' message, and logs the failed request for review.		
TC-15	Item dimensions exceed maximum bin size (invalid input)	1. Enter item dimensions greater than the largest configured bin size. 2. Trigger allocation.	Item volume > max bin capacity	System rejects allocation with 'Item Exceeds Bin Capacity' error and suggests manual/oversize storage handling.	Not Executed	Pending
TC-16	AI allocation service times out or is unreachable	1. Simulate AI/ML microservice downtime. 2. Trigger a new item allocation request.	AI service unresponsive (timeout > 5 sec)	System falls back to default rule-based allocation logic and logs the AI-service failure for monitoring.	Not Executed	Pending

Module 4: Inventory Discrepancy Detection & Alerts

Technique(s) applied: Decision Table Testing, Boundary Value Analysis

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-17	Scanned quantity exactly matches system quantity	1. Perform a cycle count scan of a bin. 2. Compare with system record.	Scanned qty = System qty = 30	No discrepancy alert is raised; audit log records a 'Match' entry.	Not Executed	Pending
TC-18	Discrepancy within acceptable tolerance (warning only)	1. Perform cycle count where scanned qty differs from system qty by an amount within tolerance (e.g., tolerance = 2 units).	System qty = 30, Scanned qty = 29 (diff = 1, within tolerance)	System logs a 'Minor Variance' warning but does not raise a critical alert.	Not Executed	Pending
TC-19	Discrepancy exactly equal to the alert threshold (boundary)	1. Perform cycle count where the difference equals the configured threshold exactly.	Threshold = 3 units; System qty = 30, Scanned qty = 27 (diff = 3)	System triggers a discrepancy alert (boundary condition confirmed to trip the rule).	Not Executed	Pending
TC-20	Sensor reports a negative or invalid quantity reading	1. Simulate a faulty RFID reader returning a negative count.	Scanned qty = -5 (invalid)	System rejects the reading, flags it as 'Invalid Sensor Data', and does not update inventory.	Not Executed	Pending
TC-21	Discrepancy detected while central system is temporarily down	1. Simulate central server downtime. 2. Perform a scan resulting in a discrepancy. 3. Restore system connectivity.	System offline during discrepancy event	Alert is generated locally, queued, and successfully delivered once system connectivity is restored; no alert is lost.	Not Executed	Pending

Module 5: Order Fulfillment & AMR Routing

Technique(s) applied: State Transition Testing, Decision Table Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-22	Order progresses through valid lifecycle states	1. Create a new order. 2. Verify transition Pending → Picking → Packed → Shipped.	New order with 3 valid line items	Each state transition occurs only in the correct sequence; timestamps recorded for each state change.	Not Executed	Pending
TC-23	AMR is assigned and completes a picking task	1. Order enters 'Picking' state. 2. AMR is auto-dispatched to the bin location. 3. AMR retrieves item and returns to pack station.	Valid pick task with reachable bin location	AMR navigates without collision, retrieves the correct item, and order moves to 'Packed' state.	Not Executed	Pending
TC-24	Last available unit of an item is picked (boundary)	1. Set item stock to 1 unit. 2. Process an order picking that unit.	Stock qty = 1 (boundary)	Stock updates to 0, item is marked 'Out of Stock', and reorder alert is generated.	Not Executed	Pending
TC-25	Attempt to fulfill an order for an out-of-stock item (invalid)	1. Set item stock to 0. 2. Attempt to create/process an order for that item.	Stock qty = 0	System blocks the pick task, notifies the user 'Item Out of Stock', and does not dispatch an AMR.	Not Executed	Pending
TC-26	AMR encounters low battery or obstacle mid-route (exceptional)	1. Dispatch AMR on a picking route. 2. Simulate low battery (<10%) or an obstacle mid-route.	Battery < 10% or obstacle detected	AMR reroutes or returns to the charging dock; the task is automatically reassigned to another available AMR without order loss.	Not Executed	Pending
TC-27	Order cancelled while in 'Picking' state (exceptional)	1. Create an order and move it to 'Picking'. 2. Cancel the order before AMR completes the task.	Order status = Picking, cancellation requested	AMR halts current pick, returns the item to its bin, and order transitions to 'Cancelled' state; inventory count restored.	Not Executed	Pending

Module 6: Warehouse Performance Reporting

Technique(s) applied: Equivalence Partitioning, Boundary Value Analysis

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-28	Generate a standard daily performance report	1. Login as Manager. 2. Select 'Generate Report' for a date with normal activity. 3. Export report.	Date = a day with recorded transactions	Report generates with accurate throughput, accuracy %, and utilization metrics; export succeeds (PDF/Excel).	Not Executed	Pending
TC-29	Generate report for a date range with zero recorded transactions (boundary)	1. Select a date range with no warehouse activity. 2. Generate report.	Date range with 0 transactions	Report generates successfully showing zero values in all metrics; system does not crash or show an error.	Not Executed	Pending
TC-30	Generate report with an invalid date range (invalid input)	1. Select an end date earlier than the start date. 2. Attempt to generate report.	Start date > End date	System displays a validation error 'Invalid Date Range' and does not generate the report.	Not Executed	Pending

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-31	Generate report during peak load with a large dataset (exceptional/performance)	1. Trigger report generation for a full quarter of high-volume data during peak system usage.	~1 million transaction records	Report completes within the defined SLA (e.g., ≤ 30 seconds) without timeout or system degradation.	Not Executed	Pending

Module 7: User Authentication & Role-Based Access

Technique(s) applied: Boundary Value Analysis, Equivalence Partitioning

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-32	Valid login with correct Admin credentials	1. Enter valid username/password for an Admin account. 2. Click Login.	Valid Admin credentials	User is authenticated and granted full access to all dashboard modules.	Not Executed	Pending
TC-33	Warehouse Staff attempts to access Admin-only configuration screen	1. Login as Warehouse Staff. 2. Attempt to navigate directly to the Admin configuration URL/module.	Role = Warehouse Staff	Access is denied with 'Insufficient Permissions' message; action is logged.	Not Executed	Pending
TC-34	Repeated failed login attempts trigger account lockout (exceptional)	1. Enter an incorrect password 3 consecutive times for the same account.	3 consecutive invalid password attempts	Account is locked after the 3rd failed attempt; user is notified and required to reset password or wait a cooldown period.	Not Executed	Pending
TC-35	Password field boundary length validation	1. Attempt registration/password change using passwords of 7, 8, 20, and 21 characters.	Passwords: 7, 8, 20, 21 chars (min=8, max=20)	7 and 21 characters are rejected; 8 and 20 characters are accepted (boundary values pass).	Not Executed	Pending

Module 8: IoT Environmental Monitoring

Technique(s) applied: Boundary Value Analysis, Exceptional Case Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-36	Temperature reading within the safe configured range	1. Simulate sensor reading within normal range. 2. Observe dashboard.	Temp = 22°C (range 15–25°C)	No alert raised; reading logged as normal.	Not Executed	Pending
TC-37	Temperature reading exceeds threshold by a small margin (boundary)	1. Simulate sensor reading at threshold + 1°C.	Temp = 26°C (threshold = 25°C)	System immediately raises a temperature-excursion alert to designated staff.	Not Executed	Pending

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-38	IoT sensor goes offline (exceptional)	1. Simulate sensor disconnection/power loss. 2. Observe system behaviour over the next monitoring cycle.	Sensor heartbeat not received for > 5 minutes	System flags sensor status as 'Offline', displays last known reading with a timestamp, and notifies maintenance staff.	Not Executed	Pending

Module 9: Non-Functional Requirements

Technique(s) applied: Performance, Security, Usability & Reliability Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-39	RFID scan-to-inventory-update latency under normal load	1. Perform a standard scan under normal system load. 2. Measure time from scan event to inventory update.	Normal load (baseline traffic)	Update reflected in system within ≤ 2 seconds (per NFR1).	Not Executed	Pending
TC-40	System handles concurrent RFID reader load (scalability)	1. Simulate 500 concurrent RFID reader input streams. 2. Monitor system response time and error rate.	500 concurrent readers, sustained 10 min	System processes all events without crash; response time remains within acceptable SLA; error rate < 0.1%.	Not Executed	Pending
TC-41	Unauthorized API access attempt is rejected (security)	1. Call a protected inventory API endpoint without a valid auth token.	Missing/invalid API token	Request is rejected with HTTP 401 Unauthorized; attempt is logged for security audit.	Not Executed	Pending
TC-42	New user completes a basic inventory search task (usability)	1. Provide a first-time user with the dashboard and a task: 'Find current stock of Item X'. 2. Measure time and steps taken, without prior training.	First-time user, standard dashboard	Task is completed within 3 clicks and under 1 minute, confirming intuitive usability (per NFR4).	Not Executed	Pending
TC-43	System recovers automatically after a simulated server failure (reliability)	1. Simulate an abrupt application server restart during active operations. 2. Observe recovery behaviour.	Simulated crash/restart	System auto-recovers, resumes processing queued events, and restores full functionality within the defined RTO (e.g., ≤ 5 minutes); no data loss.	Not Executed	Pending

6. Requirement Traceability / Test Coverage Matrix

The matrix below maps each functional and non-functional requirement to the test case(s) that verify it, confirming that all identified requirements have adequate test coverage.

Requirement	Covered By Test Case(s)
FR-1 (RFID Registration)	TC-01, TC-02, TC-03, TC-04, TC-05

Requirement	Covered By Test Case(s)
FR-2 (Real-Time Update)	TC-06, TC-07, TC-08, TC-09, TC-10, TC-11
FR-3 (AI Storage Allocation)	TC-12, TC-13, TC-14, TC-15, TC-16
FR-4 (Discrepancy Detection)	TC-17, TC-18, TC-19, TC-20, TC-21
FR-5 (Order Fulfillment / AMR)	TC-22, TC-23, TC-24, TC-25, TC-26, TC-27
FR-6 (Performance Reporting)	TC-28, TC-29, TC-30, TC-31
FR-7 (Auth & RBAC)	TC-32, TC-33, TC-34, TC-35
FR-8 (IoT Monitoring)	TC-36, TC-37, TC-38
FR-9 (Manual Override / Audit)	TC-09, TC-20 (covers correction & flagging paths)
FR-10 (Movement History)	TC-06, TC-07, TC-22 (movement logging verified)
NFR-1 (Performance)	TC-39
NFR-2 (Scalability)	TC-40
NFR-3 (Reliability/Availability)	TC-11, TC-21, TC-43
NFR-4 (Usability)	TC-42
NFR-5 (Security)	TC-34, TC-41
NFR-6 (Accuracy)	TC-17, TC-18, TC-19
NFR-7 (Interoperability)	TC-41 (API access validated)
NFR-8 (Maintainability)	TC-16 (fallback/modular behaviour)

7. Conclusion

This Test Case Design document identifies 10 functional and 8 non-functional requirements for the Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform, decomposed into 12 test scenarios and 43 detailed test cases across 9 modules. Test cases were designed using Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing to ensure coverage of normal, boundary, invalid, and exceptional conditions. The accompanying traceability matrix confirms that every identified requirement is covered by at least one test case, providing a solid foundation for the test execution phase.